

SIVARAMARAJU KALIDINDI

Senior SDET • Test Automation Engineer • QA Framework Architect

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PROFESSIONAL SUMMARY

Senior SDET and Test Automation Engineer with 3+ years architecting enterprise-grade **continuous quality engineering** platforms across financial services, healthcare, and EdTech — cutting regression cycle time by **60%** and defect leakage by **40%**. SME for **Core-Prime**, a production Selenium-Playwright Hybrid Framework adopted as the company-wide automation standard, integrating Selenium WebDriver BiDi, ThreadLocal parallel execution, Java 21 OOP with **Factory/Strategy/Builder/Singleton/Observer design patterns**, and a 10-datastore DB validation matrix.

Deep end-to-end STLC expertise: functional, regression, smoke/sanity, UAT, integration, and **JMeter performance testing** — CI/CD delivery via Jenkins/GitLab/Gradle parameterised across DEV→SIT→UAT. Currently integrating AI-augmented capabilities (RAG pipelines, LLM-based failure analysis, MCP integration) for next-generation quality engineering. Brings a mentoring track record, Agile-native delivery discipline, and measurable quality outcomes aligned to enterprise-scale transformation.

TECHNICAL SKILLS

Languages:	Java 21 (Advanced — OOP, Design Patterns: Factory, Singleton, Strategy, Builder, Observer), SQL, Gherkin; Python (scripting/data validation), JavaScript (basic)
Automation Frameworks:	Selenium WebDriver, Selenium IDE, Selenium Grid, Playwright, Selenium BiDi (WebDriver BiDi), TestNG, JUnit, Cucumber BDD/TDD, POM, Hybrid Dual-Engine Framework, DataFaker
API Testing:	RestAssured, Karate (awareness), Postman, Swagger, SOAP UI (exposure), REST/SOAP, JSON/XML, JSON schema validation, SLA response-time assertions, contract testing, end-to-end hybrid workflow validation
Testing Types:	Functional, Regression, Smoke/Sanity, UAT, Integration, Cross-Browser, Parallel Execution, Performance, API Automation, ETL/Data Validation, Manual (foundational)
Frameworks & Build:	Spring Context, Spring Profiles, Apache DBCP2, Gradle (strict dependency locks, background config cache), Maven
CI/CD & DevOps:	Jenkins (parallel worker nodes, parameterized pipelines), GitHub Actions, GitLab CI (exposure), Git, GitHub, Docker (familiarity), DEV→SIT→UAT pipeline promotion
Databases & Data:	Oracle, Snowflake, MySQL, PostgreSQL, MongoDB, Redis, DynamoDB; JDBC, SQL (JOINS, GROUP BY, window functions, aggregations), ETL pipeline testing, Apache DBCP2 connection pooling
Performance Testing:	JMeter — Load, Stress, Spike, Throughput; bottleneck diagnosis; root cause analysis; CI/CD pipeline integration
Reporting & Observability:	Allure (video capture, step screenshots), ExtentReports (CSS/memory-optimised, Base64 failure logs), Cucumber dual-listener event binding, defect leakage tracking
IDEs & Tooling:	IntelliJ IDEA, Eclipse, VS Code, JIRA, Confluence, GitHub Copilot
AI-Augmented Testing:	RAG pipeline integration (LangChain, Qdrant, OpenAI embeddings), MCP server integration, LLM-based failure triage piped to Allure, SLM on-device test generation, Healenium self-healing locators — implementing on FHLBNY AHP project (AWS deployment, Q3 2026)
Process & Methodology:	Agile/Scrum, SDLC/STLC, Test Planning, Risk-Based Testing, Code Review, Sprint Retrospectives, Global Delivery

PROFESSIONAL EXPERIENCE

Senior Software Quality Engineer (SDET)

Prospect Infosystem — Client: Federal Bank of New York | Hyderabad | Feb 2024 – Present

- Architected **Core-Prime**, a production-grade Hybrid Test Automation Framework (Selenium-Java 21, OOP, Factory/Strategy/Builder design patterns, TestNG, Cucumber BDD, Selenium Grid, Allure) end-to-end — raised test coverage **35%**, eliminated **8+ hrs/week** of manual regression, accelerated release frequency **30%** through Jenkins CI/CD; adopted as the company-wide automation **SME standard** across all active projects with zero test rewrites.
- Engineered a Selenium-Playwright Hybrid Bridge — a unified driver abstraction layer auto-delegating to Playwright for unsupported operations or executions exceeding 15-second thresholds — enabling **100% thread-safe parallel cross-browser execution** (Chrome, Firefox, Edge) with zero test rewrites on migration.

- Implemented **Selenium WebDriver BiDi** (WebSocket bidirectional protocol) for real-time network interception, browser log capture, and DOM event monitoring — replacing non-standard CDP workarounds with spec-compliant, future-proof browser communication; enabled financial-grade network traffic assertions directly within test execution.
- Designed and executed end-to-end test plans covering **functional, regression, smoke/sanity, UAT, integration, and cross-browser** suites across Credit & Deposit Administration, Discount Window Lending, and Community Lending modules — aligned to STLC milestones and Agile sprint cadences.
- Delivered **RestAssured API automation** validating BI dashboard metrics against Snowflake SQL aggregations across REST/JSON endpoints — cut defect leakage **40%** and enabled zero-regression BI releases verified against Swagger API contracts and Postman regression collections.
- Authored complex multi-table SQL JOIN queries (aggregations, GROUP BY, window functions) from developer schemas; executed cross-platform DB validation Oracle↔Snowflake via JDBC — detecting transformation errors, row-count mismatches, and null propagation across data pipelines.
- Validated Tidal-scheduled ETL pipeline jobs end-to-end through risk-based test prioritisation — improved data integrity **35%** and reclaimed **10 hrs/week** from manual reconciliation; coordinated findings with data engineering teams in Agile sprint reviews.
- Implemented **JMeter** load and stress test suite — conducted root cause analysis on a critical throughput bottleneck pre-release, improving system throughput **45%** and reducing API response time by **250 ms**; integrated performance module into Jenkins CI/CD for continuous SLA monitoring.
- Scaled parallel DB validation via Selenium Grid with isolated JDBC connections per thread asserting UI state against Oracle/Snowflake concurrently — delivering an additional **40% coverage increase** and halving validation cycle time.
- Mentored **3 junior engineers** on POM architecture, Cucumber BDD design, SQL validation, and code review best practices — structured knowledge transfer reduced onboarding time from 3 weeks to **1 week** and elevated sprint delivery quality across all team modules.
- Drove code review participation across automation PRs — conducted **root cause analysis** on selector failures, FluentWait anti-patterns, and assertion gaps; enforced framework standards that reduced flaky test rate by **60%** over two sprint cycles, eliminating pre-release emergency patching.

AI-Augmented Test Intelligence — Affordable Housing Program (AHP) | FHLBNY

- Designing an AI-augmented test intelligence layer for the FHLBNY Affordable Housing Program — architecting **MCP (Model Context Protocol)** server integration, a **RAG pipeline** over test suite documentation, and **LLM-based failure analysis** to auto-triage Allure report defects; targeting AWS-hosted deployment within the FHLBNY secure financial network (pending client access approval, Q3 2026).
- Evaluating **SLM (small language model)** integration for on-device test code generation and self-healing locator suggestions — benchmarking against Healenium baselines to quantify execution speed improvement and reduction in flaky test maintenance overhead; demo environment in active build.
- Establishing AI-assisted code quality standards for the AHP automation suite — using LLM-based static analysis to enforce selector strategy, FluentWait patterns, and assertion completeness at PR review stage; projected to reduce manual code review overhead by **40%** per sprint cycle.

Automation Test Engineer

Sadup Softech Pvt Ltd | Hyderabad | May 2023 – Dec 2023

- Designed a comprehensive test strategy for **MySchoolBot ERP** covering functional, regression, smoke, and integration test suites across 6 modules — established the first structured STLC process on the project.
- Automated critical regression paths using **Selenium + Java + TestNG** — reduced manual regression effort **40%** and compressed the sprint-over-sprint regression cycle from 2 days to **4 hours**, directly enabling same-week UAT sign-offs.
- Validated fee and attendance module submissions against backend DB via multi-table SQL JOINS — ensured end-to-end data integrity for **5,000+ student records** per release, catching 3 critical data-mapping defects pre-UAT.
- Participated in daily Agile/Scrum standups, sprint planning, and retrospectives — collaborated with developers on defect triaging and test case refinement, reducing defect re-open rate by **25%** through precise acceptance criteria alignment.
- Developed module-grouped **ExtentReports** with step-level screenshots and client-tailored failure summaries across 4 portals — enabled non-technical stakeholders to triage defects independently, saving **2+ hrs/week** of QA-to-client clarification effort.

QA Trainee — Manual Testing & Test Design

Sadup Softech Pvt Ltd | Hyderabad | Nov 2022 – Apr 2023

- Designed and maintained **200+ test cases** (positive, negative, boundary-value, equivalence partitioning) for KareClouds Healthcare SaaS platform — organised into modular regression and smoke suites subsequently adopted as the baseline library for the automation

team.

- Identified, documented, and lifecycle-tracked **60+ critical defects** in JIRA through root cause analysis and reproducible reproduction steps — reduced average developer resolution time by **24 hours** and drove a **0% defect escape rate** to UAT across two release cycles.
- Authored multi-table SQL JOIN queries across patient, appointment, billing, and clinical record tables — validated HIPAA-sensitive data handling, confirmed zero PII exposure in API responses, logs, and exports, and produced compliance evidence consumed by the delivery manager.
- Proactively upskilled from manual QA to Selenium-Java automation within **5 months** — self-studied POM, TestNG, and JDBC, securing promotion to Automation Test Engineer ahead of the planned 12-month track.

KEY PROJECTS

Core-Prime — Hybrid Test Automation Framework | github.com/sivaramaraju-k/Core-Prime

Stack: Java 21 · Selenium WebDriver · Selenium Grid · Playwright · Selenium BiDi · TestNG · Cucumber BDD · POM · Spring Context · Gradle · RestAssured · DataFaker · Allure · ExtentReports · Jenkins · MySQL · PostgreSQL · Oracle · Snowflake · MongoDB · Redis · DynamoDB

- **Thread-Safe Hybrid Architecture:** Dual-engine framework enabling runtime switching between Selenium and Playwright via centralized environment properties — 100% thread-safe parallel execution using ThreadLocal contexts across drivers, API clients, and data factories.
- **Enterprise Database Validation Matrix:** Spring Profiles + Apache DBCP2 supporting dynamic runtime connection switching across 10+ relational and NoSQL datastores (Oracle, Snowflake, MySQL, PostgreSQL, MongoDB, Redis, DynamoDB); seamless UI-to-DB assertions and ETL integrity checks.
- **API & Performance SLA Integration:** RestAssured wrapper with SLA response-time validation, JSON schema matching, and ThreadLocal auth state management — enables hybrid workflow: trigger UI action → verify API payload → assert DB record in a single end-to-end test.
- **Selenium BiDi & Local Infrastructure Orchestration:** WebDriver BiDi for WebSocket-based real-time network interception, DOM event monitoring, and JS exception capture; custom ProcessBuilder module autonomously spins up/tears down local deployment servers from TestNG @BeforeSuite.
- **Unified Dual-Reporting Dashboard:** Cucumber BDD scenarios bound to both Allure and ExtentReports via asynchronous event listeners — custom CSS + memory optimisations render massive JSON payloads and Base64-embedded failure screenshots without browser lag.
- **Resilience & Test Architecture:** Smart Retry + Dynamic Wait engine using TestNG RetryAnalyzer and FluentWait/WebDriverWait with zero Thread.sleep() calls — 100% CI pipeline stability across all parallel execution environments.

EDUCATION

B.Tech — Electrical & Electronics Engineering

Sagi Rama Krishnam Raju Engineering College, Bhimavaram, AP | Sep 2022

CERTIFICATIONS

- ISTQB Foundation Level (CTFL) — In Progress | Exam Targeted: 2025
- Udemy: Selenium WebDriver with Java — Basics to Advanced + Frameworks (Rahul Shetty Academy)
- Udemy: Cucumber BDD with Java — MasterClass (Rahul Shetty Academy)

PROFESSIONAL DEVELOPMENT

Career Transition: Self-transitioned into Software Quality Engineering from EEE — progressed QA Trainee → Automation Engineer → SDET within 2 years through structured self-study in Java, Selenium, and enterprise test automation.

Current AI Upskilling: AI-Augmented Test Architect stack — Healenium (AI self-healing locators), LLM-based failure analysis piped to Allure, RAG on test suites, Appium 2.0 (mobile), k6/Gatling (performance), Kubernetes (container orchestration for test infra), advanced design patterns (Factory, Singleton, Strategy, Builder, Observer, Chain of Responsibility).

Domain Expertise: Financial Services (Federal Banking, credit/deposit administration, ETL pipelines, BI dashboards), Healthcare (HIPAA compliance, PII validation, clinical data), EdTech (ERP, attendance and fee management systems).

Languages: English (Professional), Telugu (Native), Hindi (Conversational).